Causal Inference Crash Course:

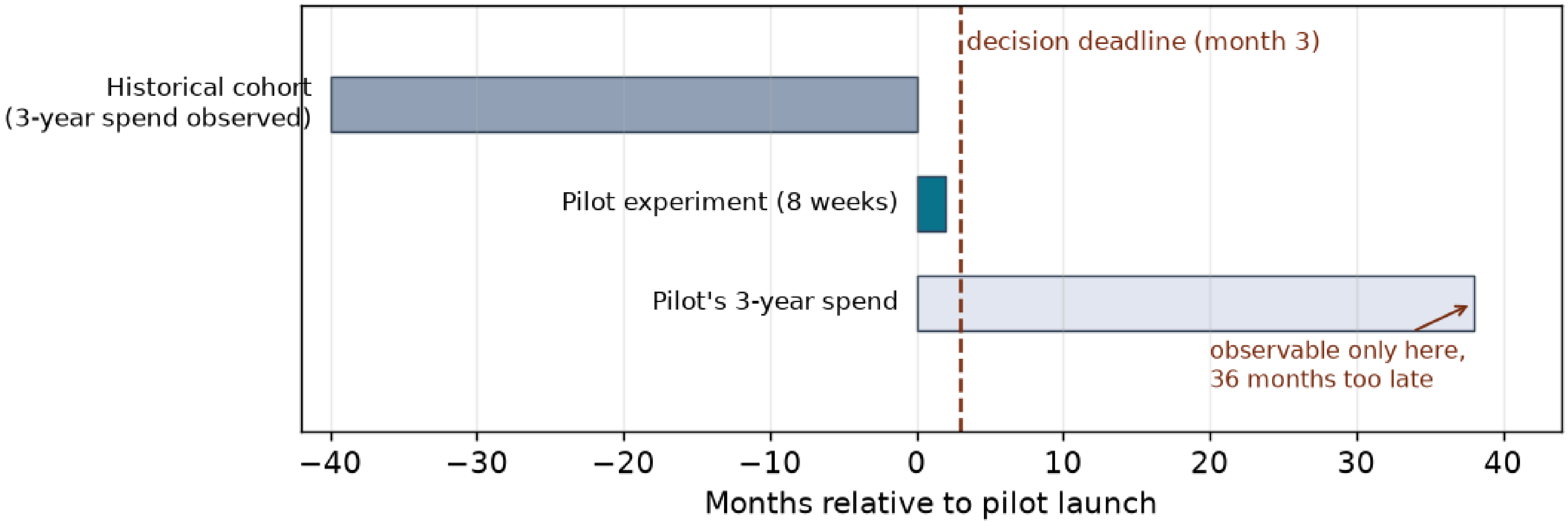
Surrogate Models

Julian Hsu

Overview

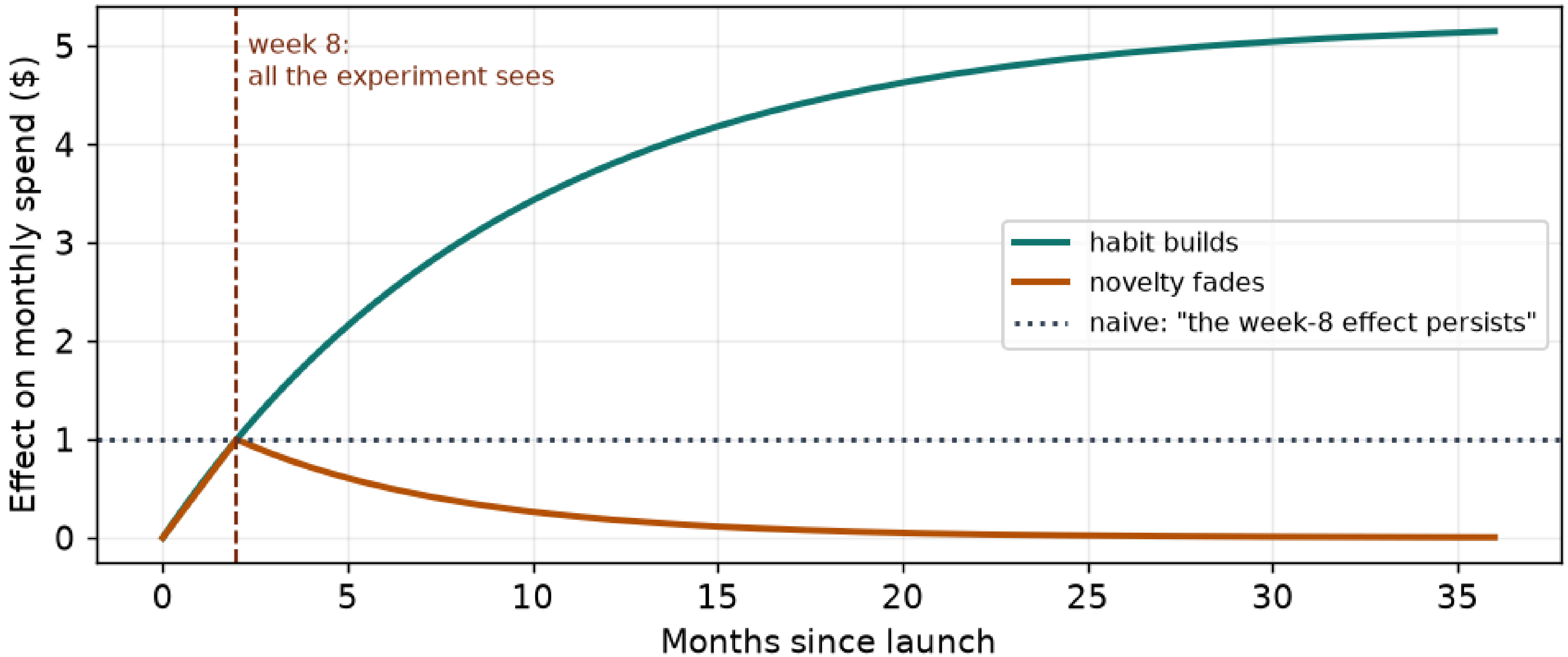
- **Surrogate models**: estimate the effect on a **long-term outcome** before that outcome exists, by bridging through intermediate ("surrogate") outcomes.
- Earlier decks asked *is the effect identified?* Here the treatment is randomized — identification is free — but the outcome won't arrive until **after the decision deadline**.
- We cover:
    - The waiting problem and the big idea
    - The surrogate index: two samples, three steps
    - The three assumptions, and what breaks
    - Diagnostics you can actually run
    - A simulated worked example
- Assumes the earlier decks: potential outcomes, randomized experiments, OLS, inference.

The waiting problem



- We pilot a **loyalty program** (W_i, randomized 50/50) for 8 weeks; the metric that matters is **3-year customer spend** (Y_i).
- An 8-week pilot must justify a 3-year claim by month 3 — waiting 36 months for Y is an experiment design, not a decision strategy.

Two tempting shortcuts



- **Declare on the short-run metric**: assume the week-8 winner is the 3-year winner.
- **Extrapolate**: assume the week-8 effect persists.

The Big Idea

- The treatment can only change the long run **through things it changes now**.
- Call those intermediate outcomes **surrogates** S_i: week-8 spend, purchase frequency, engagement, …
- Two facts we *can* measure by month 3:
  - The **experiment** tells us how treatment moves the surrogates ($W \to S$).
  - **History** tells us how surrogates map into long-term outcomes ($S \to Y$).
- **Chain them**: $W \to S \to Y$.
- One clinical surrogate: Prentice (1989). Many surrogates combined into an **index**: Athey, Chetty, Imbens & Kang (2019).

The surrogate index

Two samples, three steps, one prediction model.

Two samples

Sample	W (treatment)	X (baseline)	S (week-8)	Y (3-year)
Experimental — the pilot	✓	✓	✓	*not yet*
Observational — historical cohort	—	✓	✓	✓

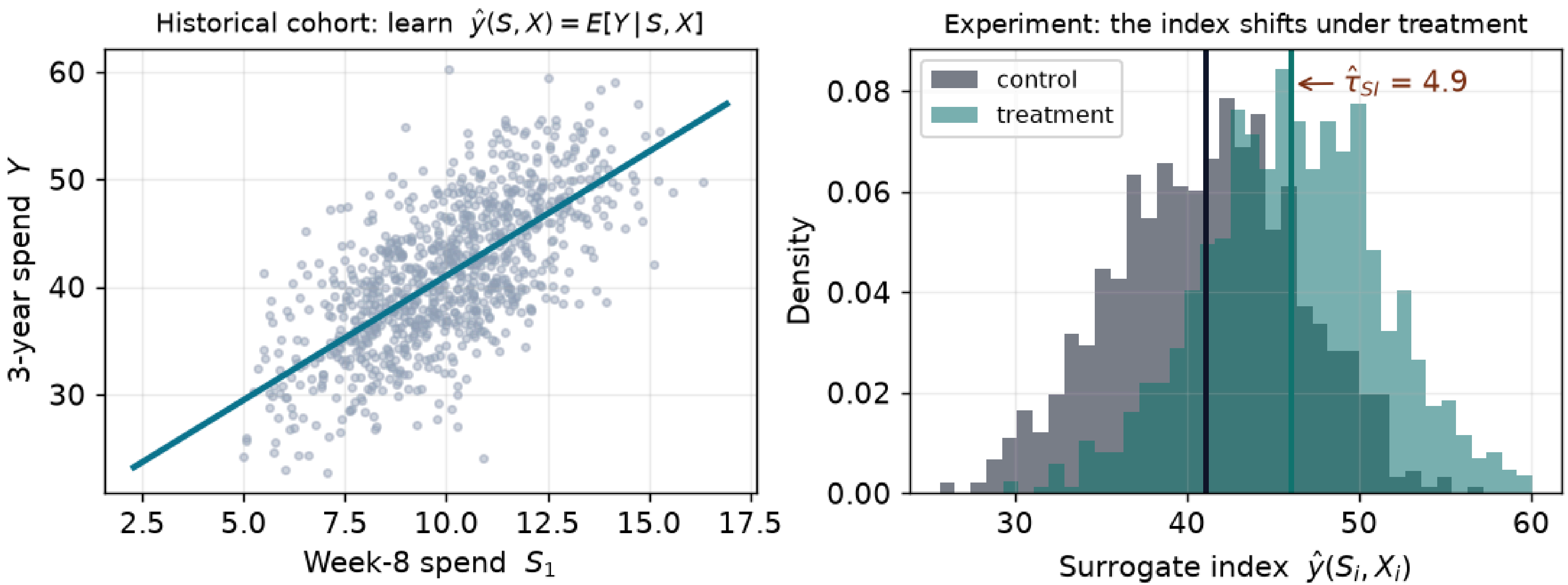
- The historical cohort predates the program: those customers' full arcs — week-8 behavior *and* 3-year spend — are already on file.
- The pilot has the causal contrast; history has the horizon. Neither alone answers the question.

Three steps

1. **Learn the bridge** (observational sample): fit $\hat{y}(s,x) = \hat{E}[Y \mid S=s, X=x]$.
    - A pure prediction task — OLS here, but any model that predicts well works.
2. **Score the pilot** (experimental sample): compute the **surrogate index** $\hat{Y}_i^{SI} = \hat{y}(S_i, X_i)$ — each customer's week-8 behavior converted into predicted 3-year spend.
3. **Difference it**:

$$\hat{\tau}_{SI} = \frac{1}{n_1}\sum_{i:\,W_i=1}\hat{Y}_i^{SI} - \frac{1}{n_0}\sum_{i:\,W_i=0}\hat{Y}_i^{SI}$$

What it looks like



- Left: history traces the $S \to Y$ map.
- Right: treatment shifts the surrogate distribution; pushed through the map, the index shifts with it. The gap between arm means is $\hat{\tau}_{SI}$.

Why it works

- Randomization gives $\tau = E[Y_i \mid W_i = 1] - E[Y_i \mid W_i = 0]$ — but Y is missing in the pilot.
- Iterated expectations: $E[Y \mid W = w] = E\big[\, E[Y \mid S, X, W = w] \,\big|\, W = w\big]$.
- **Surrogacy** deletes the inner W: $E[Y \mid S, X, W] = E[Y \mid S, X]$.
- **Comparability** lets history estimate that object: it is the same $E[Y \mid S, X]$ the observational sample identifies — our $\hat{y}(s, x)$.
- So averaging $\hat{y}(S_i, X_i)$ by arm recovers each arm's mean of Y — without ever observing Y in the pilot.

Assumptions

Three. The first is the one that should keep you up at night.

Assumption 1: surrogacy

$$W_i \perp Y_i \mid S_i, X_i$$

- Given the surrogates (and baseline), treatment status carries **no extra information** about the long-run outcome: every causal path from W to Y passes through S.
- For a *single* surrogate this is the **Prentice criterion** — and it is almost always false.
- The modern move: make S a **rich bundle**. The more of the treatment's short-run footprint you record, the less room for a bypassing channel.
- Untestable at the horizon you care about — and a merely *predictive* surrogate is not enough: see **Appendix — the surrogate paradox**.

Assumption 2: comparability

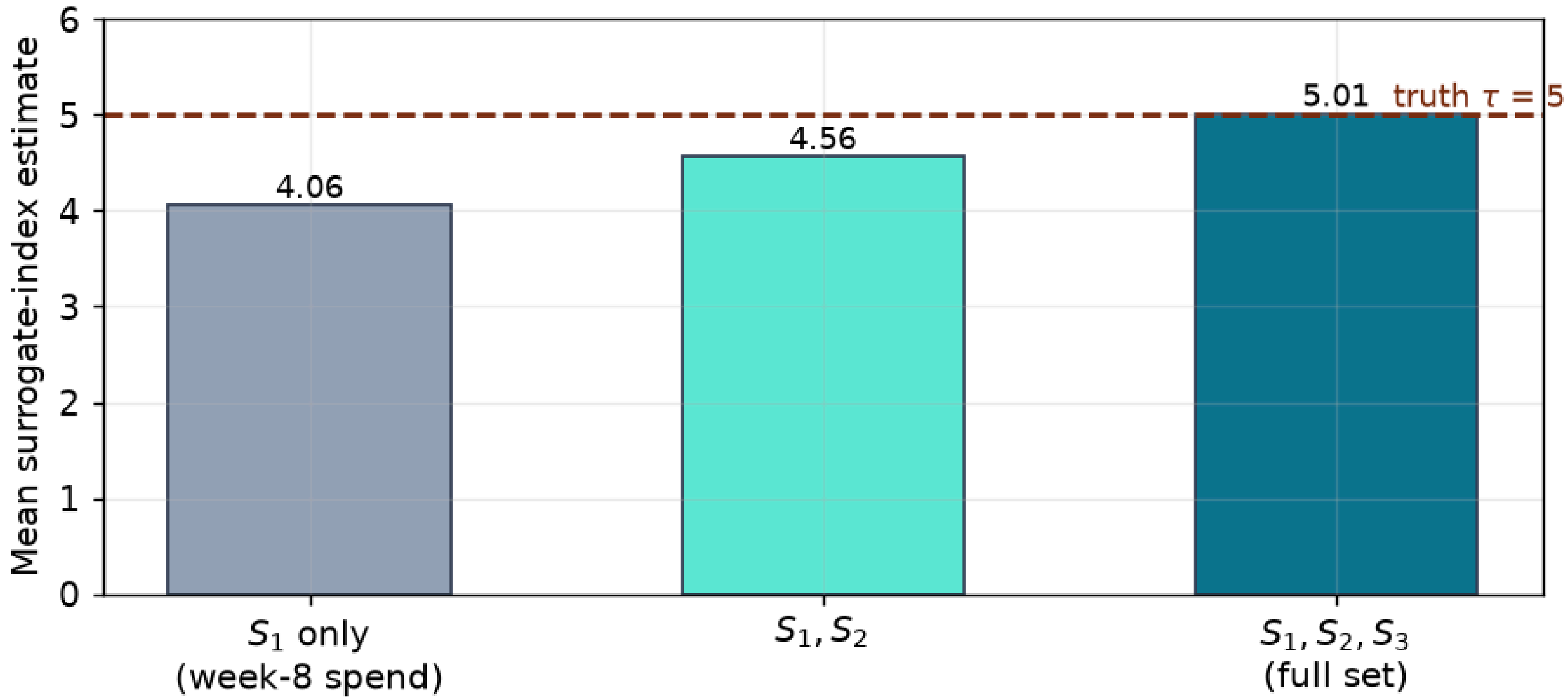
- The $S \to Y$ map must be the **same** in both samples: $Y_i \perp G_i \mid S_i, X_i$, where G_i flags which sample unit i belongs to.
- Threats:
    - **Different populations** — history covers all customers, the pilot only app users.
    - **Different regimes** — history spans a boom, the pilot runs into a recession.
    - **The program rewires the map** — pull-forward: promo-induced week-8 spend that cannibalizes year-2 spend predicts Y differently than organic week-8 spend does.
- Plus **overlap**: the pilot's (S, X) values must actually occur in history — the bridge cannot be extrapolated.

Assumption 3: a clean experiment

- $W_i \perp \big(Y_i(1), Y_i(0), S_i(1), S_i(0)\big)$ — free when the pilot is randomized; in an observational "pilot" you are stacking unconfoundedness (deck 2) *on top of* surrogacy.

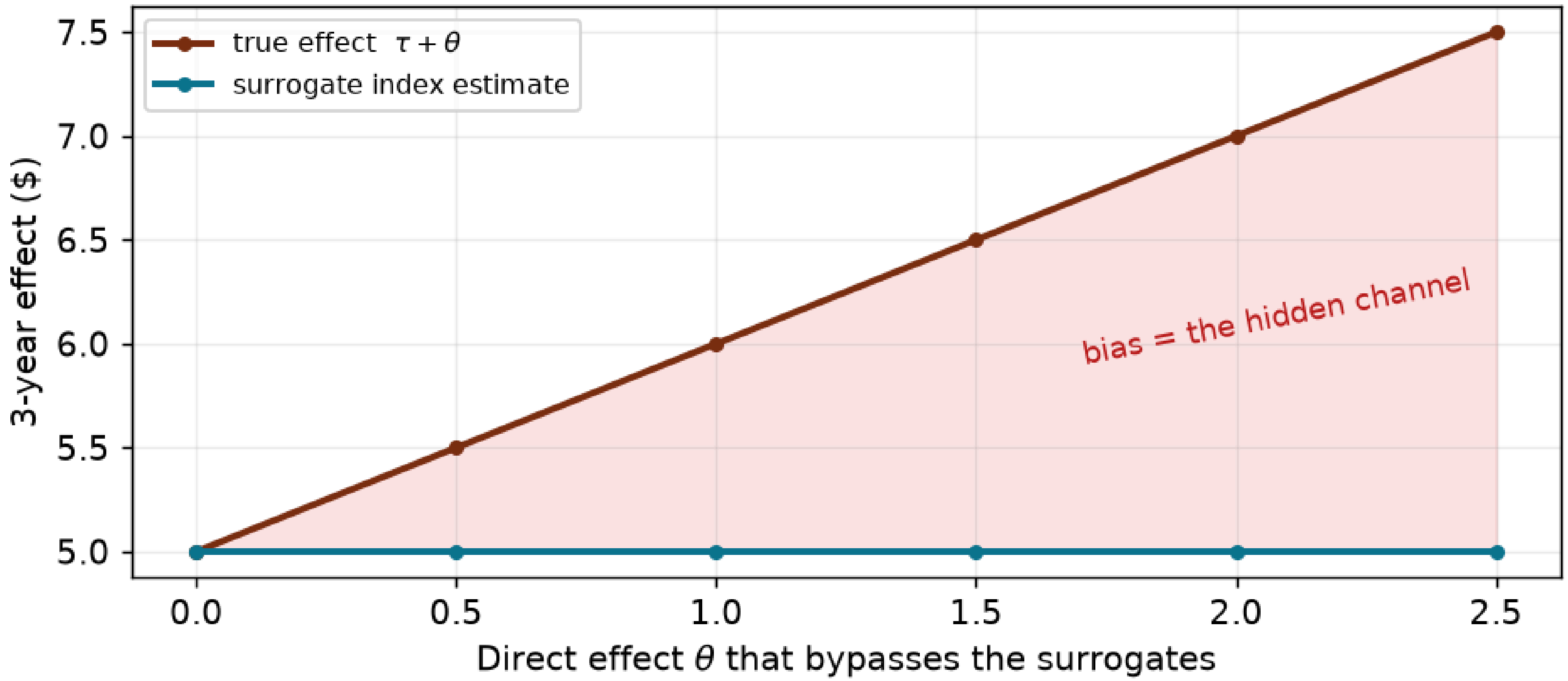
Assumption	What it buys	Main threat
1. Surrogacy	W drops out given (S, X)	a channel no surrogate records
2. Comparability	history estimates the map	population / regime shift, pull-forward
3. Clean experiment	the causal $W \to S$ contrast	rarely an issue in a true A/B test

One surrogate is fragile



- Each surrogate alone carries only part of the effect; the estimate climbs toward the truth as the bundle fills out.
- This is why Athey et al. call it a surrogate **index** — their application bridges through *several quarters* of employment history, not one number.

When surrogacy fails



- Give the treatment a channel no surrogate records (size θ): the index misses **exactly** that channel — bias $= \theta$, silently, with tight confidence intervals around the wrong number.

- The direction is reasoned about, not tested: a missed channel working *with* the treatment means the index **understates**; against, **overstates**. Athey et al. (2019) derive bounds under partial surrogacy.

Diagnostics

- **Backtest on finished experiments**: for past A/B tests whose long-run outcomes have since matured, compute the week-8 index and compare it to the realized effect. *The* industry validation.
- **Horizon laddering**: validate the 8-week index against 6-month outcomes (already observable today), the 6-month index against 1-year, … — surrogacy failures usually surface early.
- **Leave-one-surrogate-out**: if dropping any single surrogate moves $\hat{\tau}_{SI}$ materially, the bundle is too thin to trust.
- **Re-fit the bridge on the pilot's control arm** as its early outcomes trickle in and compare with history's bridge — a direct comparability check.

Worked example

Simulated pilot, so we know the answer ($\tau = 5$).

The setup

- Pilot: $n = 2{,}000$ customers randomized 50/50, run for 8 weeks. Historical cohort: $n = 10{,}000$ customers with 3-year spend on file.
- Three week-8 surrogates: **early spend**, **purchase frequency**, **engagement**; one baseline covariate (prior spend). True 3-year effect $\tau = 5$.
- Two scenarios:
  1. **Valid** — the treatment reaches 3-year spend *only* through the three surrogates.
  2. **Hidden channel** — an extra effect of $\theta = 1.5$ bypasses them (true effect $= 6.5$).
- Companion notebook with all code here.

Four estimators

1. **Short-run difference**: the arm gap in week-8 spend, reported as if it were the 3-year effect.
2. **Single surrogate**: the surrogate index built from week-8 spend alone (Prentice-style).
3. **Surrogate index**: the index from all three surrogates plus the baseline covariate (OLS bridge).
4. **Oracle — wait 3 years**: the pilot's actual long-run difference. The benchmark the deadline forbids.

Bias

- Mean bias $(\hat{\tau} - \tau)$ over 500 simulations:

Estimator	Valid ($\tau = 5$)	Hidden channel ($\tau = 6.5$)
Short-run difference	-3.00	-4.50
Single surrogate	-0.98	-2.48
Surrogate index	-0.01	-1.51
Oracle (wait 3 years)	+0.00	+0.00

- With a rich bundle and valid surrogacy the index is clean; under the hidden channel it is off by **exactly** $\theta = 1.5$.
- One-draw picture with confidence intervals: **Appendix — the estimates, drawn**.

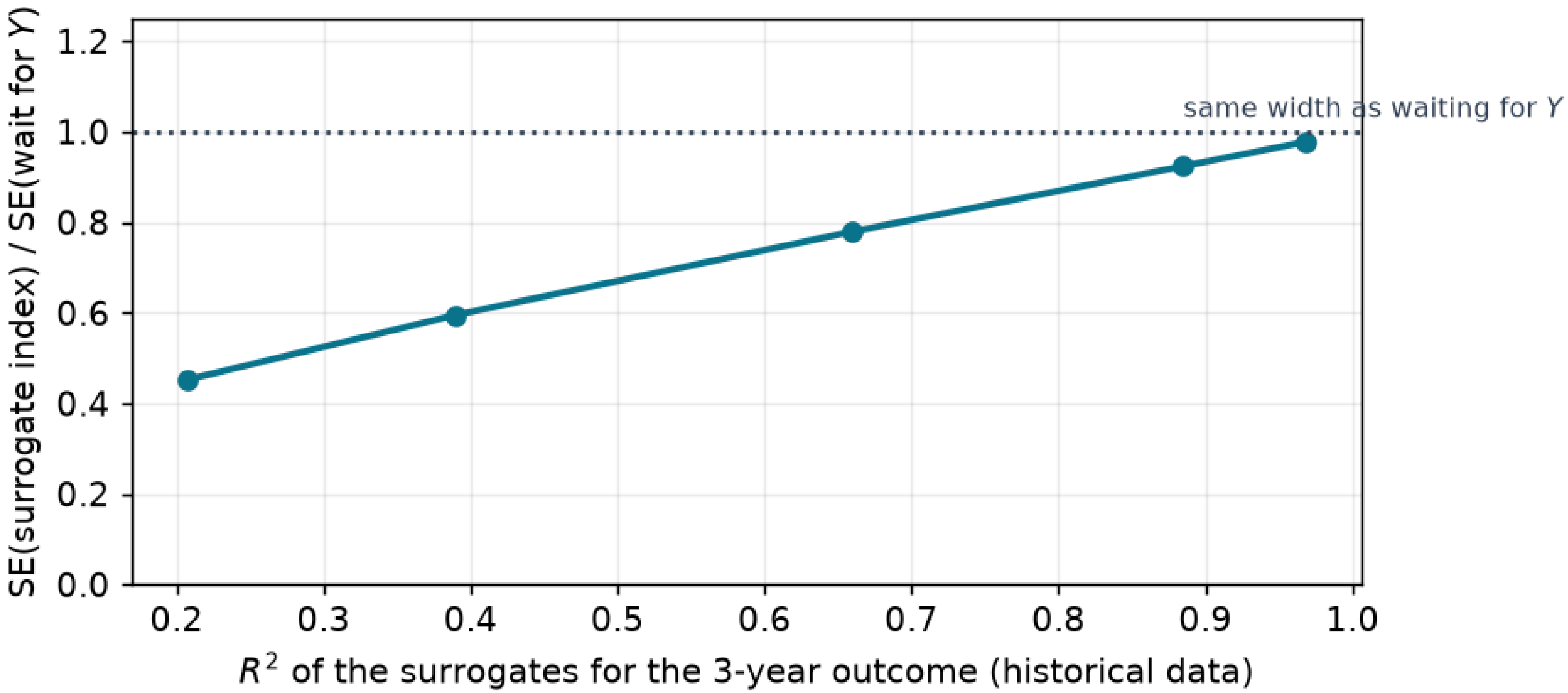
Inference: the index is a prediction, not data

- Two stages carry noise: the experiment *and* the estimated bridge $\hat{y}(\cdot)$.
- A **plug-in** CI (t-test treating the index as data) ignores stage 1; the **bootstrap** resamples both samples and refits the bridge every draw.

95% CI for the surrogate index	Coverage, $n_{obs} = 10{,}000$	Coverage, $n_{obs} = 500$
Plug-in	0.95	0.85
Bootstrap (both stages)	0.94	0.91

- A large historical sample makes the shortcut nearly harmless; a thin one flatters your precision. Athey et al. (2019) provide analytic (influence-function) standard errors.

A precision bonus



- In the worked example the index's SD (0.25) *beats* the oracle's (0.31): the index averages away the part of Y that surrogates cannot predict, while waiting keeps all of that noise.

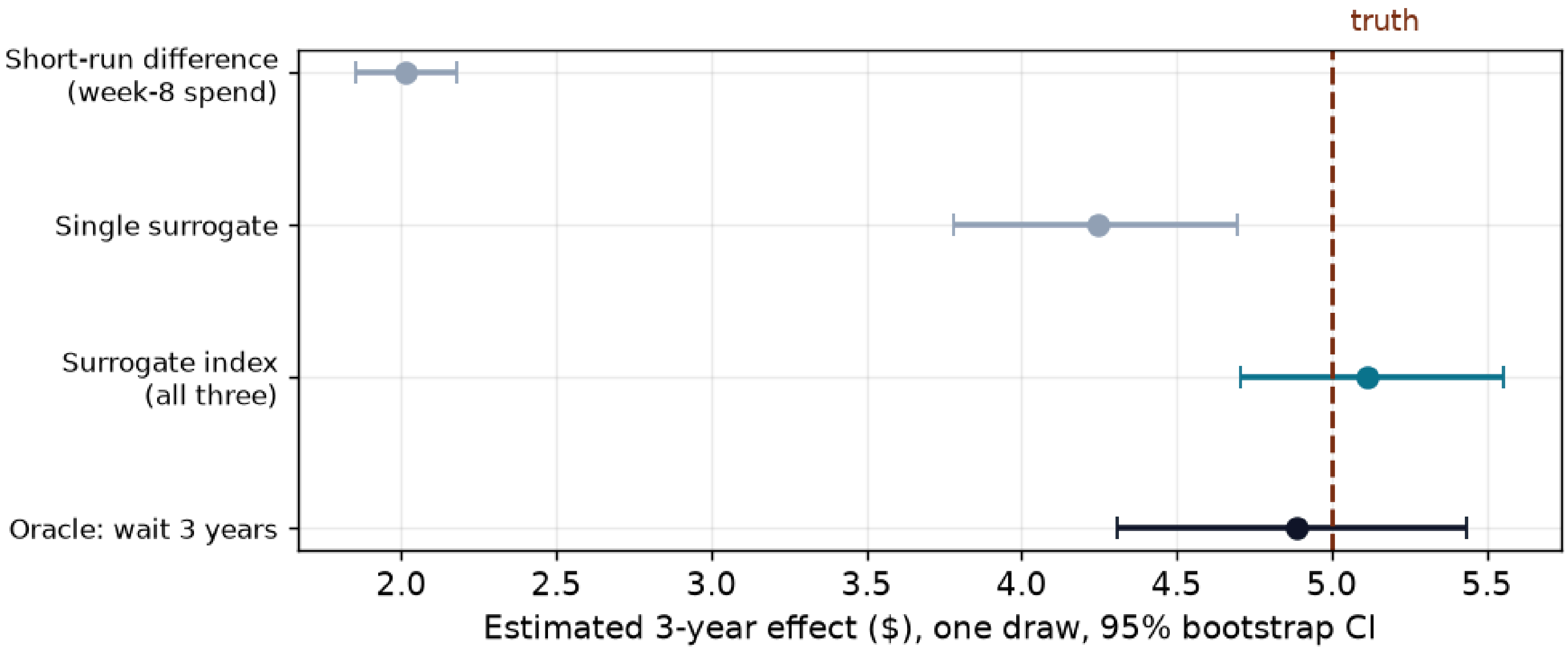
- The lower the surrogates' R^2 for Y, the bigger the precision advantage — and the more nervous surrogacy should make you. Precision is not validity.

Conclusion

- The surrogate index chains the two things you can measure by the deadline: the experiment's $W \to S$ and history's $S \to Y$.
- Pros: an answer **when the decision is due**; stage 1 is ordinary supervised prediction; often *tighter* CIs than waiting for the outcome.
- Cons: surrogacy is untestable at the horizon that matters; a bypassed channel biases silently; comparability and overlap constrain which history you may borrow.
- Reach for it when the decision can't wait — then **backtest against finished experiments** and keep logging richer surrogates.

Appendix Slides

Appendix: the estimates, drawn



- One draw of the worked example (valid scenario), 95% bootstrap CIs.
- The short-run difference and the single surrogate are *precisely wrong* — tight intervals far from the truth. The index and the oracle both cover it; the index is a bit tighter.

Appendix: the surrogate score

- The index is one of **two mirror-image estimators** in Athey et al. (2019).
- **Surrogate index**: work in the *experimental* sample, replacing missing Y with predicted $\hat{y}(S, X)$.
- **Surrogate score**: work in the *observational* sample, where Y is real, weighting each unit by how "treated" its surrogate bundle looks:

  $$\hat{q}(s, x) = \hat{P}(W = 1 \mid S = s, X = x) \text{ estimated in the experimental sample,}$$

  then average observational Y_i with weights $\propto \hat{q}(S_i, X_i)$ for the treated mean and $\propto 1 - \hat{q}(S_i, X_i)$ for the control mean — a propensity-style reweighting where surrogates play the role of covariates.
- Combining both nuisance models yields a **doubly-robust** estimator: consistent if *either* the bridge or the score is well specified, with better efficiency (Chen & Ritzwoller 2023; Kallus & Mao 2020).

Appendix: bias under partial surrogacy

- Decompose the true effect as $\tau = \tau_S + \theta$: the part transmitted through recorded surrogates, plus the bypass.
- Under comparability and a clean experiment, $E[\hat{\tau}_{SI}] = \tau_S$, so

$$\mathrm{bias}(\hat{\tau}_{SI}) = -\theta.$$

- Reasoning about θ:
    - Sign: does the unrecorded channel plausibly work with or against the recorded ones?
    - Size: Athey et al. (2019) bound the bias using the treatment's residual association with Y given S — the smaller the room left after conditioning on a rich S, the tighter the bound.
- Practical translation: every additional orthogonal surrogate you log shrinks the worst case.

Appendix: the surrogate paradox

- It is possible for the treatment to **raise every surrogate**, every surrogate to **positively predict** Y — and the treatment to **lower** Y (VanderWeele 2013).
- How: an unobserved variable raises S but lowers Y, or a bypassing negative direct effect — predictive correlation survives, the causal chain does not.
- The cautionary classic: in the CAST trial, anti-arrhythmic drugs suppressed arrhythmias (the accepted surrogate for cardiac death) and *increased* mortality.
- Moral: a surrogate that merely *predicts* is not a surrogate that *transmits*. Surrogacy is the conditional-independence statement $W \perp Y \mid S, X$ — check paths, not correlations.

Appendix: surrogates vs. mediation vs. principal stratification

- **Mediation analysis** decomposes an observed total effect into channels; it needs sequential ignorability *and* you already have Y. Different goal: explanation, not forecasting.
- **Principal stratification** (Frangakis & Rubin 2002) defines "principal surrogacy" via joint potential values of S — conceptually cleaner, much harder to estimate.
- The **surrogate index** claims less than either: it never interprets the $S \to Y$ coefficients causally — the bridge is *only a prediction model*. What it must swallow instead is total mediation: **no bypass at all**.
- Corollary: don't read $\hat{y}$'s coefficients as "the value of engagement"; they are predictive weights, nothing more.

Appendix: practical guidance

- Log **many, cheap, hard-to-game** surrogates: breadth of behavior (frequency, recency, category breadth, engagement, support contacts) beats one clever metric.
- **Pre-register the bridge model** — sample, features, algorithm — before unblinding the pilot, or the index becomes a metric-shopping exercise.
- Beware **Goodhart's law**: once teams target the index, the historical $S \to Y$ map is exactly what breaks (a comparability violation you caused).
- Maintain a **backtest library**: every past experiment whose long-run outcomes have matured is a free validation point for the current index.
- Refresh the bridge on rolling cohorts; regime changes (pandemic, price overhaul, new product mix) invalidate old maps.

Literature Review of Related Papers

- **Foundational (biostatistics)**
  - Prentice (1989). "Surrogate endpoints in clinical trials: Definition and operational criteria." *Statistics in Medicine.* link
  - Freedman, Graubard, Schatzkin (1992). "Statistical validation of intermediate endpoints for chronic diseases." *Statistics in Medicine.* link
  - Frangakis & Rubin (2002). "Principal Stratification in Causal Inference." *Biometrics.* link
  - VanderWeele (2013). "Surrogate measures and consistent surrogates." *Biometrics.* link
- **The surrogate index & econometrics**
  - Athey, Chetty, Imbens, Kang (2019). "The Surrogate Index: Combining Short-Term Proxies to Estimate Long-Term Treatment Effects More Rapidly and Precisely." NBER WP 26463. link

- Athey, Chetty, Imbens (2020). "Combining Experimental and Observational Data to Estimate Treatment Effects on Long Term Outcomes." link
- Kallus & Mao (2020). "On the role of surrogates in the efficient estimation of treatment effects with limited outcome data." link
- Chen & Ritzwoller (2023). "Semiparametric Estimation of Long-Term Treatment Effects." *Journal of Econometrics.* link

- **Applications & industry practice**

- Hohnhold, O'Brien, Tang (2015). "Focusing on the Long-term: It's Good for Users and Business." *KDD.* link
- Yang, Eckles, Dhillon, Aral (2024). "Targeting for Long-Term Outcomes." *Management Science.* link